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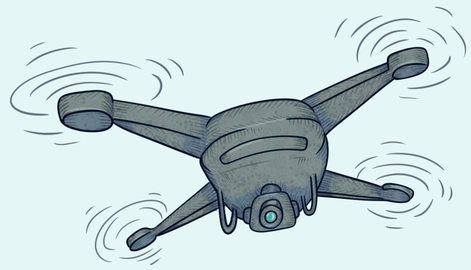
PORPOISE MONITORING USING A PROGRAMMED UAV

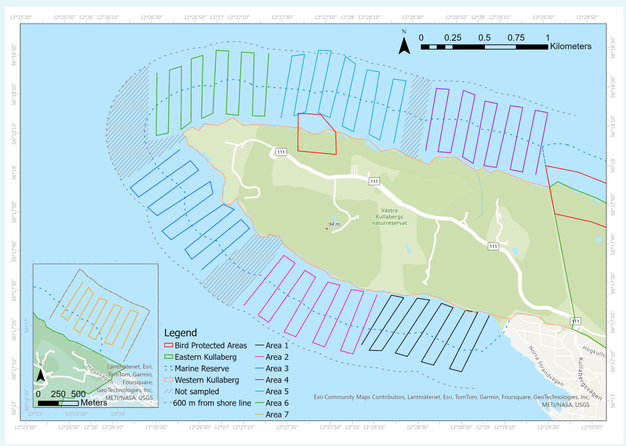
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BACKGROUND

- Standardised monitoring is an essential tool in the conservation of many protected species.
- Elusive marine mammals can be challenging to study via traditional visual methods.
- We explore the use of UAV transect monitoring for investigation into the density and distribution of harbour porpoises (*Phocoena phocoena)* within a restricted high-density protected area for the species in Sweden.





UAV TRANSECT MONITORING METHOD

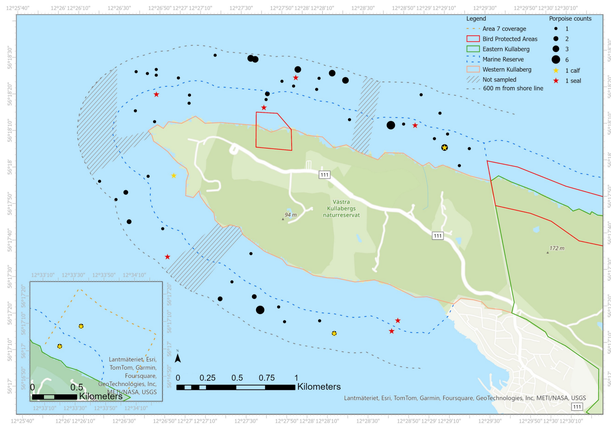
- Visual data collected by UAV (DJI Phantom 4 Pro) twice weekly during 17 weeks in June to September 2023.
- Pre-programmed transects distanced 100 m apart.
- UAV flight speed of 10 m/s and camera FOV 55 x 105 m at 66 m altitude.



ANALYSIS

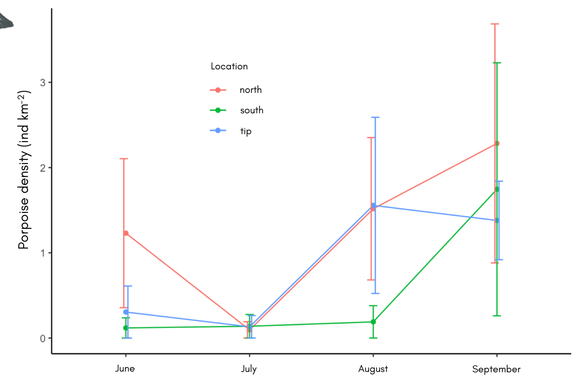
- Manual analysis of all videos twice in normal speed by same person.
- 25% of videos reviewed by second independent non-expert observer.
- Statistical analysis to investigate density and evaluate method performance.

RESULTS



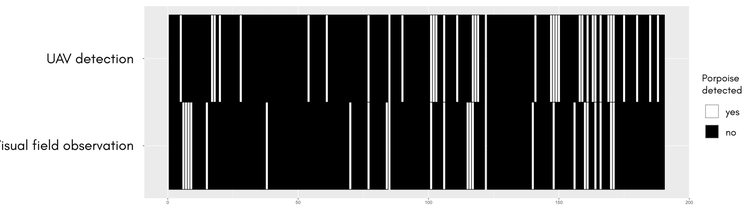
Porpoise density increased from June to September.

Spatial differences in porpoise density on micro-scale.

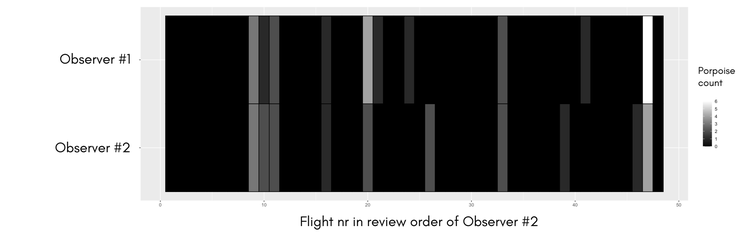


Site	Area monitored per flight (km^{-2})	Total nr of flights	Total area monitored (km^{-2})	Total nr of porpoises	Mean density (ind km^{-2})	Mean density 95% CI (ind km^{-2})
Area 1	0.53	29	15.37	3	0.20	0.00 - 0.49
Area 2	0.45	29	13.05	14	1.28	0.00 - 3.29
Area 3	0.47	29	13.63	9	0.65	0.12 - 1.18
Area 4	0.41	29	11.89	17	1.43	0.00 - 3.10
Area 5	0.52	29	15.08	21	1.38	0.41 - 2.35
Area 6	0.41	28	11.48	10	1.00	0.10 - 1.91
Area 7 (reference)	0.52	15	7.8	4	0.51	0.00 - 1.25
Total	3.31	188	88.3	78	0.95	0.51 - 1.40

More porpoises detected with UAV-method compared to traditional visual method:



No significant observer bias (W = 26.5, p = 0.67):



UAV TRANSECT MONITORING SHOW GREAT POTENTIAL FOR STANDARDISED DATA COLLECTION OF MARINE MEGAFAUNA IN SPECIFIC AREAS OF INTEREST.

CONCLUSIONS

The UAV transect monitoring method show great potential for standardised data collection, especially at small spatial scale, as:

- It can be modified and applied to a variety of research fields and species.
- It is easily implemented and presents logistic and economic benefits.
- Data collection and video review does not require experienced observers with biological expertise.
- Perception bias, observer bias, fatigue and learning effects can be reduced, prevented and quantified.